User Manual
The ASL-DVS Suite

How to use the ASL-DVS web tools day to day: what every page, tab and button does, with a technical close to each chapter for when you want to know how it works. Written for the Pi Zero 2 W build; full-build differences are marked where they matter.

DASHBOARD SYSMON INSTMON WIFIMON 44HELPER

FILES COVERED BY THIS EDITION

TOOL	FILE	VERSION
Dashboard	asl_dvs_dashboard_pi02w_v9_3_71_13_20261007.py	9.3.71.13-pi02w
SysMon	sysmon_pi02w_v6_13_67_9_20261007.py	6.13.67.9-pi02w
44helper	asl_dvs_m17_44helper_v0_0_175_20261008.py	0.0.175
instmon	instmon_v2_38_3_20261007.py	2.38.3
wifimon	wifimon_v5_28_20261007.py	5.28
Installer	install_asl_dvs_v6_6_20261006.sh	6.6
SysMon FULL	sysmon_v6_13_74_20261007.py	6.13.74
Dashboard FULL	asl_dvs_dashboard_v9_3_73_20261006.py	9.3.73

DRAFT · PARTS I–II OF V

KD8PGK · written with Claude (Anthropic) · CC BY-NC 4.0 · github.com/kd8pgk/ASL-DVS

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How to use this manual

If you are new to the suite, read Part I first, then turn to the chapter for the tool in front of you. Every chapter after Chapter 2 follows the same order, so you always know where to look:

1. **What it's for**: what the tool or tab does, and when you would open it.
2. **Before you start**: what has to be installed, and anything that can go wrong.
3. **Tab by tab**: each card and button, and what good and bad readings look like.
4. **Common tasks**: short recipes, step by step.
5. **Under the hood**: the technical close, in a purple-labelled panel. It covers the files, services, ports and timers involved. You can skip it on a first read.

Markers used in the text

NOTE Background, or something worth knowing.

TIP A quicker or safer way to do something.

CAREFUL This changes your node, restarts something or needs a reboot.

FULL BUILD Applies to the full build (Pi 3, 4 or 5) only. Also shown as the badge **FULL**.

Commands and pictures

Commands are typed on the Pi, in an SSH session or at a keyboard and screen. Lines starting with `#` are notes. Replace `<pi-address>` with your Pi's IP address or host name.

```
# a note about the next line
sudo systemctl status sysmon
```

Pictures marked **Illustration** are drawn to show a page's layout. They use the same tab names and order as the real page, but the values in them are made up. Example identities use the placeholders `N0CALL` and `<NODE>`.

PART I

Getting started

What the suite is, what each tool is for, how to log in, and the short version of installing and updating it. The full install steps are in the install manuals in the `docs/` folder.

1. The suite at a glance
2. Installing and updating, in brief

The suite at a glance

ASL-DVS is a set of five small web tools that run on the same Raspberry Pi as your AllStarLink node. Each tool has its own page in the browser. Together they let you run, check, repair and update the node without typing commands.

1.1 What is on the node

The tools sit on top of a working AllStarLink 3 (ASL3) node with DVSwitch. It helps to know the main parts, because the tools check and talk to all of them:

PART	WHAT IT DOES
Asterisk + app_rpt	The ASL3 engine. It runs your node, links to other AllStarLink nodes over IAX2 and handles EchoLink.
Analog_Bridge	Turns the node's audio into digital voice and back again. It talks to Asterisk on a private node (1999 by convention).
MMDVM_Bridge	Connects Analog_Bridge to the digital networks: DMR directly, and System Fusion, P25, NXDN and D-STAR through their gateways.
Gateways	YSFGateway, P25Gateway, NXDNGateway and ircDDBGateway. Each one connects to that mode's reflectors or talkgroups.
USRP2M17	Bridges a second private node (1917 by convention) to an M17 reflector.

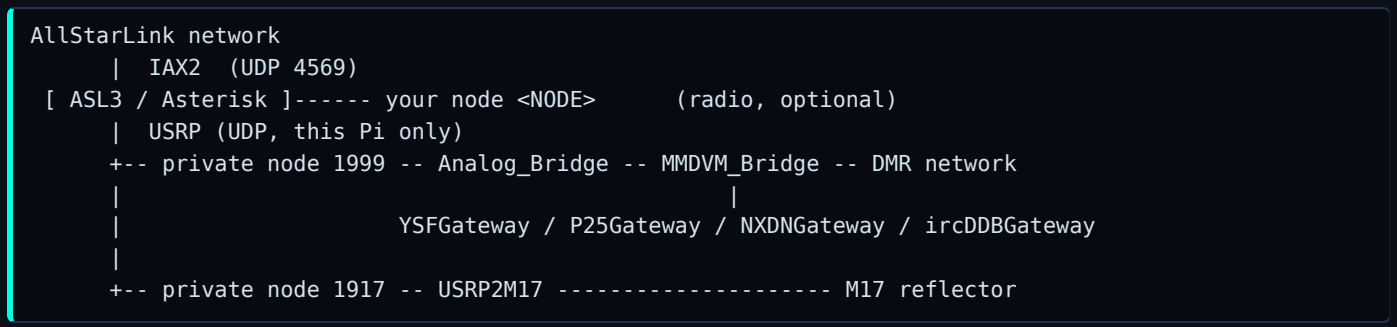


Figure 1.1 The audio path through a bridge node. Node numbers 1000 to 1999 are private: they never leave the Pi. The full layout is in `docs/ASL-DVS_Idealized_Node_Configuration.pdf`.

1.2 The five tools

Each tool is one file, runs as its own service, and serves its own page on its own port.

TOOL	PORT	WHAT IT'S FOR	OPEN IT WHEN YOU WANT TO...
Dashboard	8989	Node control.	Connect to AllStar or EchoLink nodes, change DMR talkgroup or YSF, P25, NXDN, D-STAR, XLX or M17 reflector, use favorites, make a phone call.
SysMon	9999	System monitor and health checks.	See whether the Pi and node are healthy, find a stopped service or a bad config file, read logs, look after the SD card.
44helper	9997	Setup, repair and configuration helper.	Set up 44Net or Cloudflare remote access, install ASL3, DVSwitch, SVXLink, M17 or Ampersand, run updates, save and restore your config.

instmon	8990	Installer and manager for the suite.	Update the tools from GitHub, start or stop them, pause the suite for a big job, back up or image the SD card.
wifimon	8991	WiFi and power watchdog.	Change WiFi networks, check signal and supply voltage, set up the automatic safe shutdown.

> **NOTE** 44helper is optional. The other four make up the core suite. A sixth item, the Watchdog script, is no longer needed: the Dashboard's own service restarts it if it stops.

Which tool for which job

I WANT TO...	GO TO
Talk on a DMR talkgroup or a reflector	Dashboard, the mode's tab
Link to another AllStar node	Dashboard, ASL tab
Find out why something stopped working	SysMon, Overview and Services, then the matching check tab
Check that my config files agree with each other	SysMon, ASL-DVS and DVSM tabs; 44helper, Config tab
Save my config before I change it	44helper, Config tab (restore point)
Reach my node from outside my home network	44helper, Router Install / Tunnel tabs or Cloudflare
Update the suite	instmon, GitHub Updates
Update Linux and ASL packages	44helper, Update tab
Make the SD card last longer	SysMon, SD Card tab; 44helper, System Optimization
Change WiFi network	wifimon

1.3 A first look at each page

Every page has a header with the tool's name and version and a Log Out button. Most pages have a row of tabs under the header; click a tab to switch view. The pictures below show each tool's tabs in order.

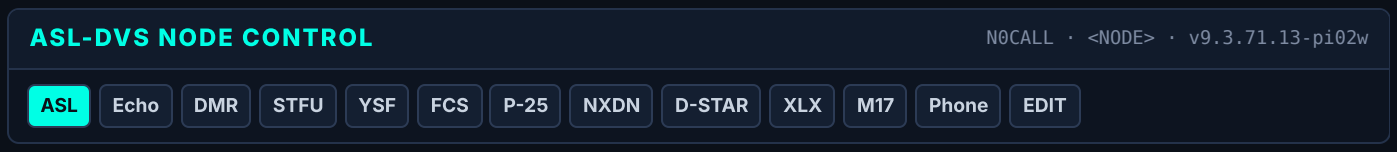


Figure 1.2 · Illustration. The Dashboard (port 8989). One tab per mode, plus Phone and EDIT. EDIT holds your favorites and settings. The Pi Zero 2 W build keeps every tab of the full build.

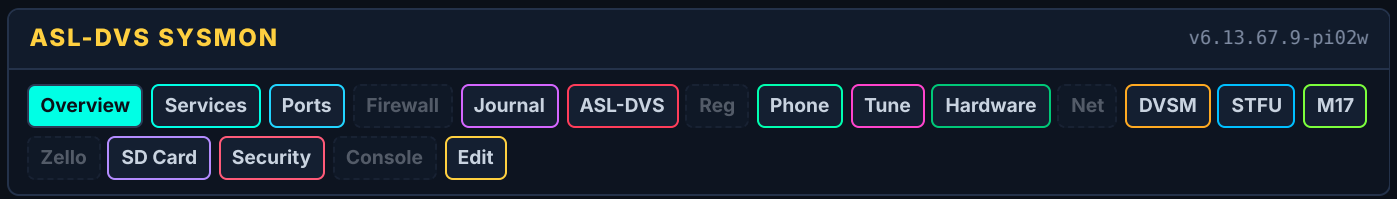


Figure 1.3 · Illustration. SysMon (port 9999). The 14 solid tabs are in the Pi Zero 2 W build. The 5 faded tabs (Firewall, Reg, Net, Zello, Console) are FULL build only.

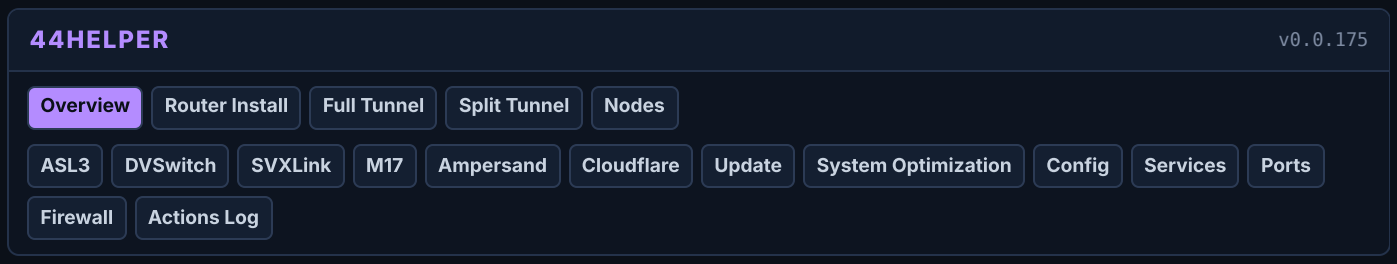


Figure 1.4 · Illustration. 44helper (port 9997). The first row is 44Net remote access. The second row has the installers, maintenance, Config and the status tabs. 44helper is the same on every Pi.

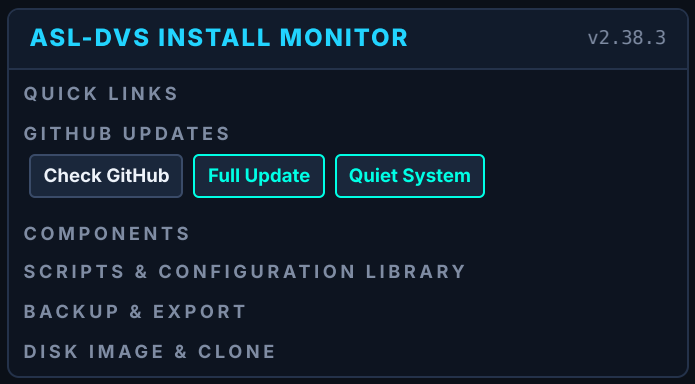


Figure 1.5 · Illustration. instmon (port 8990) is one long page of sections, not tabs.

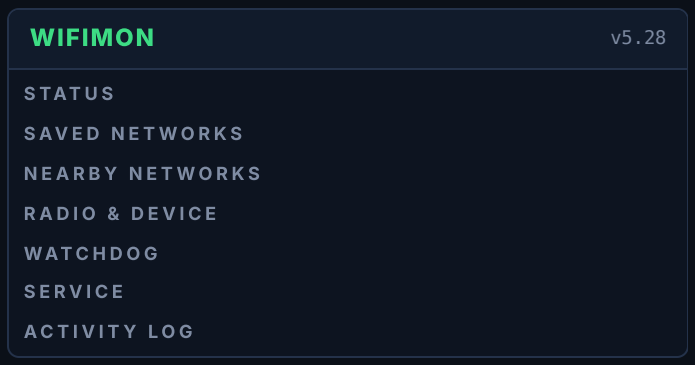


Figure 1.6 · Illustration. wifimon (port 8991) is also one page of cards.

1.4 Two builds: Pi Zero 2 W and full

The Dashboard and SysMon come in two builds. The other tools have a single build that runs on every Pi.

PI ZERO 2 W BUILD `PI02W`

For the 512 MB Pi Zero 2 W. This manual is written for it.

- Dashboard: every tab is kept. It searches the AllStar and EchoLink lists from disk instead of memory, and refreshes less often.
- SysMon: drops the Firewall, Reg, Net, Zello and Console tabs to save memory.

FULL BUILD `FULL`

For a Pi 3, 4 or 5.

- Dashboard: the same tabs, and it keeps the lists in memory.
- SysMon: all 19 tabs. Chapter 6 covers the five extra ones.

Which build am I running? Look at the version in the page header. A Pi Zero 2 W build version ends in `-pi02w`, for example `6.13.67.9-pi02w`. A full build version doesn't, for example `6.13.74`.

> **NOTE** Apart from SysMon, the builds look and work the same, so this manual points out the full build only where it differs.

1.5 Logging in

1. Open the page in a browser: `http://<pi-address>:<port>`, for example `http://192.168.1.50:9999` for SysMon.
2. Type the Pi's **root password** and press Log In. There is no user name.
3. When you're done, press **Log Out** in the header, especially on a shared computer.

- **The Dashboard and SysMon share one login.** Log in to either one and the other opens without asking, in the same browser. A shared login lasts 12 hours from the last time you used either page.
- **instmon, wifimon and 44helper each have their own login.**
- **Wrong passwords lock you out for a while.** On the Dashboard and SysMon, five wrong passwords within 5 minutes lock that address out of both for 15 minutes. The other tools have similar limits. Wait it out; don't keep guessing.
- **44helper stays logged in** until you log out, press **Log out all devices**, or the Pi reboots, because its jobs can run for a long time.

> **CAREFUL** The pages use plain HTTP, so your password crosses the network unencrypted. Only open them on your own network or over a VPN or tunnel, never straight from the internet. Chapters 7 and 8 cover safe remote access with 44helper.

> **TIP** If the login is always refused, set the root password again on the Pi with `sudo passwd root`. Raspberry Pi OS ships with root locked, so a new Pi needs this once.

1.6 Phones and tablets

The pages are laid out for phone, tablet and desktop widths. On a phone, wide tables hide their less important columns, and the tab row wraps onto more lines. Turning a tablet to landscape shows everything.

Leave a page open only while you're using it. Each open page polls its tool for fresh data, and on a Pi Zero 2 W that costs CPU and memory.

UNDER THE HOOD

Services, files and ports

TOOL	SYSTEMD SERVICE	PROGRAM	CONFIG
Dashboard	asl_dvs_dashboard	/usr/local/bin/asl_dvs_dashboard.py	/etc/asl_dvs/asl_dvs.conf
SysMon	sysmon	/usr/local/bin/sysmon.py	/etc/sysmon/sysmon.conf
44helper	44helper	/opt/44helper/asl_dvs_m17_44helper.py	/etc/44helper/44helper.conf
instmon	instmon	/usr/local/bin/instmon.py	/etc/asl_dvs/instmon_library/
wifimon	wifimon	/usr/local/bin/wifimon.py	/etc/wifimon/wifimon.conf

The Dashboard and SysMon program paths are links into `/usr/local/lib/asl_dvs/` and `/usr/local/lib/asl_dvs_sysmon/`. The installed file keeps its own name, so the version is always visible on disk.

One process each, started through the launcher

Every service starts its tool through the shared launcher `/usr/local/bin/asl_dvs_launch.py`:

```
ExecStart=/usr/bin/python3 /usr/local/bin/asl_dvs_launch.py /usr/local/bin/sysmon.py
```

The launcher loads the tool as a module so that Python keeps its compiled copy in `__pycache__`. That roughly halves each tool's memory (SysMon on a Pi Zero 2 W: 54 MB started directly, 26 MB through the launcher). The launcher also turns on three memory savers: `python3 -OO`, `MALLOC_ARENA_MAX=2` and a periodic `malloc_trim`. Each has an off-switch file in `/etc/asl_dvs/` named `launch_no_optimize`, `launch_no_arena_cap` or `launch_no_trim`.

How a page works

- Each tool is a small Python HTTP server. The page's HTML, CSS and JavaScript are built into the tool's file, so there is nothing else to install and no web server to configure.
- The page fetches live data from the tool's own `/api/...` addresses as JSON and redraws its cards.
- The login checks the password against the root entry in `/etc/shadow`, and sets a session cookie. Requests that change anything also need a matching CSRF header, so another web page can't act for you.
- Version numbers come from the `VERSION` line in each file. The `-pi02w` suffix is how the installer and instmon recognise the Pi Zero 2 W build.

Checking on a tool from the shell

```
# is it running, and its last 30 log lines
systemctl status sysmon --no-pager
journalctl -u sysmon -n 30 --no-pager
# is anything listening on its port?
ss -tlnp | grep 9999
# restart all the web tools
sudo systemctl restart sysmon asl_dvs_dashboard instmon wifimon 44helper
```

Installing and updating, in brief

This chapter covers just enough to get the suite onto a node and keep it current. For every step and option, use the install manual for your build.

MANUAL	FILE IN DOCS/
Pi Zero 2 W install	ASL-DVS_Pi02w_Install_Manual.pdf
Full build install FULL	ASL-DVS_Full_Build_Install_Manual.pdf

2.1 Before you start

- A Raspberry Pi already running AllStarLink 3 with DVSwitch set up.
- SSH access with `sudo`, and a root password (`sudo passwd root`).
- On a Pi Zero 2 W, some free memory. Check with `free -m`. The install manual lists services you can turn off if you don't use them.

2.2 Install in three steps

1. **Get the files.** Download the current files from github.com/kd8pgk/ASL-DVS into one folder on the Pi.
2. **Install instmon first.** It can then install and update everything else from the browser.

```
sudo python3 instmon_v2_38_3_20261007.py --install
```

3. **Install the rest** with the installer. On a new node it asks for your callsign, node number and a label.

```
sudo bash install_asl_dvs_v6_6_20261006.sh
```

The installer picks the build for you: the Pi Zero 2 W build on a Pi Zero 2 W, the full build on anything else. If one build is already installed, it keeps that build. When it finishes, it checks that each tool is running and listening on its port.

> **TIP** 44helper is optional. Install it the same way, or from instmon's page, when you want its remote-access, installer or Config tools.

2.3 First visit to each page

PAGE	ADDRESS	CHECK FIRST
Dashboard	http://<pi-address>:8989	Your callsign and node number are right in the header.
SysMon	http://<pi-address>:9999	The Overview shows no red items (Chapter 3).
instmon	http://<pi-address>:8990	Every tool shows as installed and running.
wifimon	http://<pi-address>:8991	Answer "Turn the watchdogs on?" (below).
44helper	http://<pi-address>:9997	Take a restore point on the Config tab (Chapter 11).

wifimon's watchdogs

A new wifimon starts with both watchdogs off: it shows WiFi and voltage, but won't act on them. On your first login it asks whether to turn them on. **Add your WiFi networks first.**

- **WiFi watchdog:** reconnects to your saved networks, and shuts the Pi down after 3 minutes with no connection.
- **Voltage watchdog:** shuts the Pi down after 30 seconds of low supply voltage.

> **CAREFUL** Both watchdogs can shut the node down. A Pi Zero 2 W is sensitive to voltage drop, so use a good power supply and a short cable before turning the voltage watchdog on.

2.4 Keeping the suite up to date

Updates come from GitHub through instmon's **GitHub Updates** section.

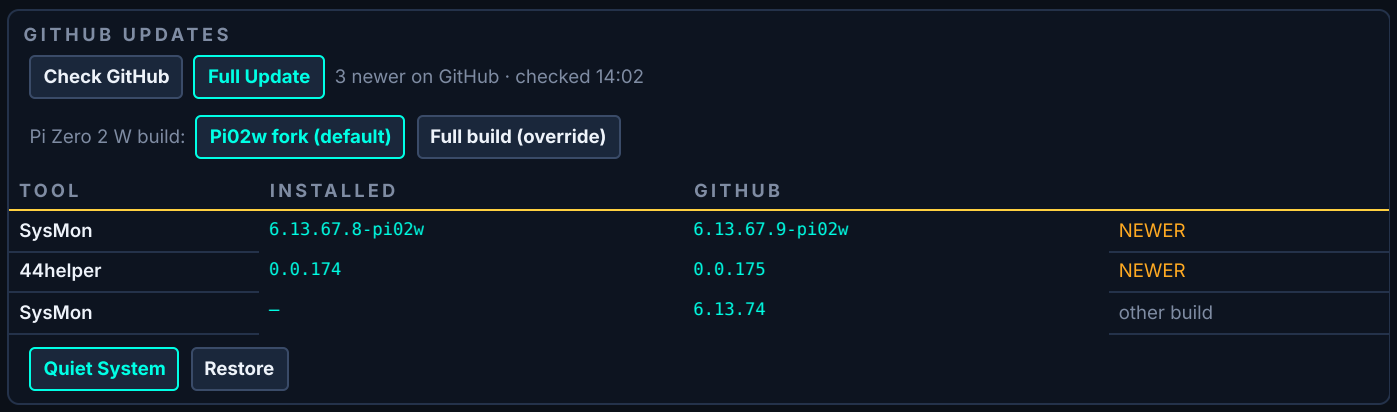


Figure 2.1 · Illustration. instmon's GitHub Updates section on a Pi Zero 2 W. The build choice appears only on a Pi Zero 2 W.

BUTTON	WHAT IT DOES
Check GitHub	Lists the newest copy of each tool on GitHub, and marks the ones newer than yours.
Update	Downloads and checks one file into instmon's library. Nothing is installed until you press Install on it.
Full Update	Updates every installed tool in one go, one tool at a time, with automatic roll-back if a new version fails.
Quiet System	Pauses SysMon, the Dashboard and 44helper during a big upload or install. The radio side keeps running, and everything comes back by itself after 30 minutes. **Restore** brings it back sooner.

> **TIP** Before a Full Update, take a restore point in 44helper's Config tab. Full Update only replaces the tools, but a restore point means you can roll back any config change you make while trying out the new versions.

Updating Linux, ASL3 and DVSwitch packages is a separate job. 44helper's Update tab handles it (Chapter 10).

2.5 Removing the suite

```
# remove one tool
sudo python3 /usr/local/bin/sysmon.py --uninstall
# remove every tool except instmon
sudo bash uninstall_asl_dvs_all_v1_3.sh
```

Your settings in `/etc/asl_dvs/`, instmon's library and wifimon's config stay in place. Remove instmon last, with its own `--uninstall`.

UNDER THE HOOD

What install and update actually do

--install, per tool

- Copies the file into place, writes its `.service` unit, and writes the shared launcher if it is missing.
- Enables the service so it starts at boot, and starts it.
- `--uninstall` stops and removes the service and the program. It removes the launcher only once no other service uses it.

How the installer picks a build

1. `SYSMON_VARIANT` / `DASH_VARIANT` set to `pi02w` or `full`, if you set them;
2. otherwise, the build already installed;
3. otherwise, `pi02w` on a Pi Zero 2 W and `full` on any other Pi.

If no file of the chosen build is found, that tool is skipped with a warning; the other build is never used in its place. The installer looks in its own folder, the current folder, `/root`, `/home/pi`, your home folder, `/tmp` and instmon's library. `--non-interactive` (or `-y`) skips every question and takes the identity from `AUTO_CALLSIGN`, `AUTO_NODE` and `AUTO_LABEL`.

The checks on every download

Before instmon saves a file from GitHub, the file must match GitHub's file list. It must also have a version line and the right first line, pass a syntax check, and support `--install`.

Full Update, step by step

1. Downloads and checks every new file before anything changes.
2. Pauses the web tools.
3. For each tool in turn, saves a copy of the installed file in the library, runs the old `--uninstall`, and runs the new `--install`. If the new one fails, it puts the old one back and moves on.
4. Updates instmon last. A timer reinstalls the old instmon if the new one isn't answering after 90 seconds.
5. Ends with a summary. New install scripts go only to the Scripts library and are never run for you.

On a Pi Zero 2 W, the build choice decides which build Full Update installs. A tool installed in the other build is listed as **SWITCH** and replaced. On other Pis the choice is hidden and each tool keeps its build.

Settings that survive an update

Config files in `/etc` are never touched by an update. SysMon's Hardware tab options (HDMI and video driver off, lower GPU memory) are saved when SysMon is uninstalled during an update and put back on install, if the install follows within 30 minutes.

PART II

SysMon

SysMon is the node's health monitor. It shows whether every service is up, checks your Asterisk, DVSwitch and M17 config files against each other, reads logs, and looks after the hardware and SD card. Most of what it does is read-only. Every button that changes something says so.

3. SysMon basics and live status
4. Node and DVSwitch health
5. System care
6. Full-build tabs

SysMon basics and live status

This chapter covers the parts of SysMon you see every time: the header and command bar, then the four status tabs: Overview, Services, Ports and Journal. Open SysMon at `http://<pi-address>:9999`.

3.1 What SysMon is for

Open SysMon when something on the node isn't working, or when you want to be sure everything is. In one place it shows:

- whether Asterisk, the DVSwitch bridges, the gateways, USRP2M17 and the ASL-DVS tools are running;
- the Pi's load, memory, temperature and power health, updated every 5 seconds;
- which programs are listening on which network ports, and whether two of them clash;
- the system log (journal) of any service, filtered and ready to copy.

The status tabs can also start, stop and restart services. Everything else in SysMon is either read-only or asks before it changes a file.

3.2 The page at a glance

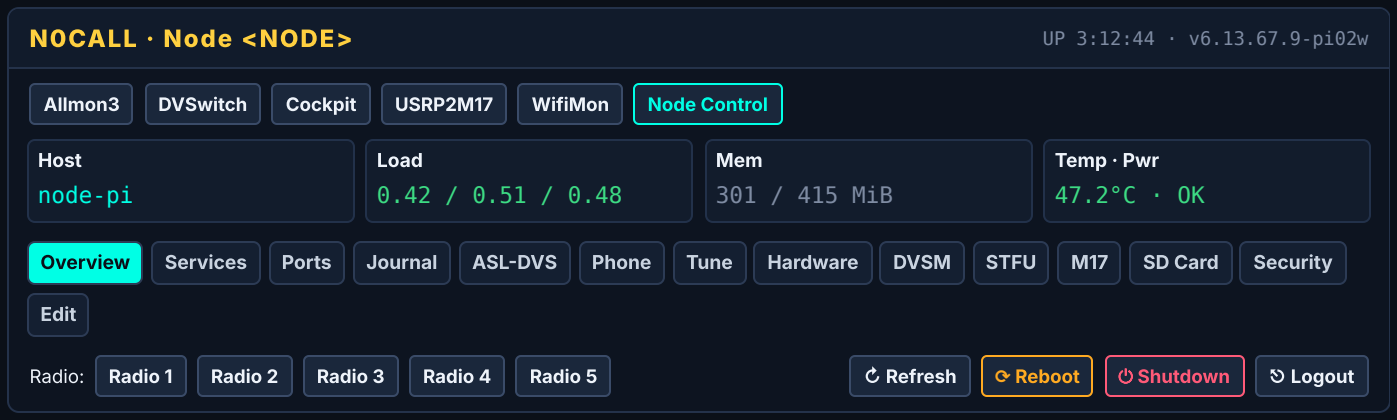


Figure 3.1 · Illustration. SysMon's top section. From the top: header, quick links, status strip, tabs and command bar.

PART	WHAT IT SHOWS OR DOES
Header	Your callsign and node number, how long the Pi has been up, and SysMon's version.
Quick links	Open Allmon3, the DVSwitch page, Cockpit, the USRP2M17 page and wifimon in a new tab. **Node Control** goes back to the Dashboard.
Status strip	Host name, load, memory, CPU temperature and power health (below).
Tabs	One per view. You can hide the ones you don't use on the Edit tab (Chapter 5).
Radio 1 to 5	Saved sets of radio audio settings. Pressing one applies it (Chapter 5, Tune tab).
Refresh	Reads the status strip again now, instead of waiting for the next 5-second update.
Reboot / Shutdown	Restart or power off the whole Pi. Both ask first. After Shutdown, the Pi stays off until you remove and restore power.

Reading the status strip

FIELD	GREEN	AMBER	RED
Load (1, 5, 15 min)	1-minute load up to 1.5	above 1.5	above 3
Temp	up to 60 °C	above 60 °C	above 75 °C
Pwr	`OK`	`WARN` : a problem happened since boot but isn't happening now	`⚡ UV` under-voltage now, or `⚡ THR` throttled now

Hover over **Pwr** to see exactly what was flagged, such as "Under-voltage since boot". When under-voltage starts, SysMon also shows a pop-up warning to check the power supply.

> **TIP** A Pi Zero 2 W has four 1 GHz cores and 512 MB, so a load of 1 to 2 during a tune or an update is normal. A load that *stays* red with nothing going on usually means a service is stuck restarting. Check the Overview for an `NR:` count.

The status dots

Every service list in SysMon uses the same coloured dots:

DOT	MEANS
● **Active**	Running normally.
● **Stopped**	Installed but not running. That is fine for a gateway you don't use, and a problem for Asterisk.
● **Failed / Masked**	It crashed or failed to start, or it has been blocked from starting (masked).
● **Unknown**	SysMon couldn't read its state.
● Not installed	There is no such service on this Pi. That is normal for optional parts.

Two banners you may see

- **"Restart sysmon to apply server address changes"**: you changed SysMon's port or bind address on the Edit tab. It takes effect when SysMon restarts.
- **"wifimon — Node Shutting Down"**: wifimon has started a safe shutdown because of low voltage or lost network. The Pi needs power-cycling, or its WiFi back, before it will come up again.

3.3 Overview tab

The Overview is SysMon's home page. It lists your **pinned services** in groups. Out of the box the groups are System, ASL-DVS, Asterisk Core and M17, which cover everything a bridge node runs.

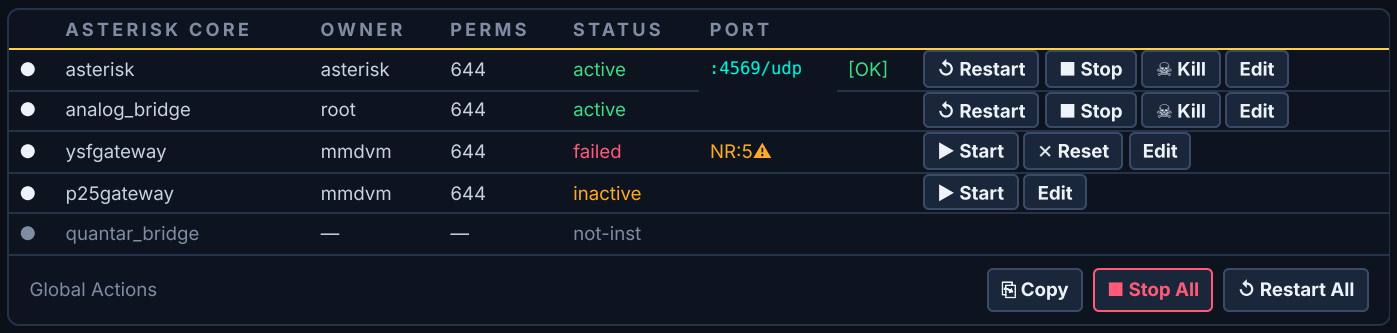


Figure 3.2 · Illustration. One Overview group. The buttons on each row depend on the service's state.

COLUMN	WHAT IT TELLS YOU
Name	The service. Click it to open its detail panel (3.4).
Owner	The Linux user it runs as. Amber means it runs as root but doesn't need to. Click the amber value for the Security tab's explanation.
Perms	The permissions on its service file. Amber or red means other users could change it. Click for the explanation.
Status	`active`, `inactive`, `failed`, `not-inst` or `unknown`.
Port	The port it should listen on. If it has restarted on its own since it was last started, this shows `NR:` and the count instead, with ⚠ above 3.
[OK] / [DUP]	For a running service with a port: `[DUP]` means another program has the same port and protocol, and hovering names both. `[OK]` means no clash.

Row buttons

STATE	BUTTONS
active	**Restart**; **Stop** (asks first); **Kill**, which ends the process at once with no clean shutdown (asks first); **Edit**, which opens the service file.
failed	**Start**; **Reset**, which clears the failed mark and the restart counter; **Edit**.
inactive	**Start**; **Edit**.

Restart is done carefully: SysMon stops the service, waits for it to stop, starts it, and waits for it to come up. The Overview refreshes itself a few times over the next half minute while that happens.

Global Actions

- **Copy** puts the whole Overview on the clipboard as text, ready to paste into a help request.
- **Stop All** stops every pinned service except SysMon itself.
- **Restart All** restarts every pinned service, SysMon included.

> **CAREFUL** **Stop All** and **Restart All** act at once, with no confirmation. Stop All takes the node off the air: it stops Asterisk, the bridges and the Dashboard. To get it back, press **Restart All** or reboot. SysMon keeps running, so the page stays usable.

3.4 Services tab

The Services tab lists **every** service on the Pi, not just the pinned ones.

- **Active** shows only running services. **All** also shows stopped, failed and disabled ones.
- Type in **filter by name** to narrow the list, for example `gateway`.
- The list refreshes every 30 seconds while the tab is open. **Copy** puts it on the clipboard as text.

The service detail panel

Click any service, here or on the Overview, to open its panel. The top shows badges for its state, whether it starts at boot, its process number, its restart count (`NR:`), CPU and memory use, and the user it runs as. Under that, **Ownership & permissions** shows who owns the service file and the program.

SECTION	BUTTON	WHAT IT DOES, IN PLAIN WORDS
Daemon Controls	Reload Daemon	Makes systemd re-read every service file. Needed after you edit one by hand.
	Reset Failed	Clears a "failed" mark so the service can be started cleanly.
	Mask / Unmask	Mask blocks a service from starting at all, even by hand, until you Unmask it. Asks first.
Runtime Controls	Start / Stop / Restart	As on the Overview. Stop asks first.
	Reload	Asks the program to re-read its settings without restarting. Greyed out for services that don't support it.
Boot Persistence	Enable / Disable	Whether the service starts by itself at boot. Disable asks first. Neither one starts or stops it now.
Pinned Services	Pin / Unpin	Adds the service to the Overview, in a group you pick or a new one, or removes it.
Status	(output)	The service's current status and its last 50 log lines, with errors in red and warnings in amber.

Editing a service file

Edit opens the service's systemd file in an editor, with **Save**, **Restart**, **Reload Daemon** and **Copy** buttons. After saving, press Reload Daemon and then Restart for the change to take effect.

> **NOTE** Service files that came with a package live in `/lib/systemd/system`. SysMon never changes those. Saving one creates your own copy in `/etc/systemd/system/`, which takes priority and survives package updates. A yellow bar warns you when this happens.

3.5 Ports tab

The Ports tab lists every network port a program on the Pi is listening on. It refreshes every 15 seconds.

COLUMN	MEANING
Dot	Green: the port belongs to a known service. Grey: a listener SysMon can't tie to a service.
Port / Proto	The port number, and TCP or UDP.
Process / PID	The program listening, and its process number.
Known Service	The service that program belongs to, or "— unknown —".
Conflict	`[OK]`, or `[DUP]` if more than one program has the port.

- **All / TCP / UDP** filter the list. **Copy table** copies what you see; **Copy all** copies every port.
- Click a row for its panel: the listen address, **Probe** buttons (HTTP and TCP for a TCP port, UDP for a UDP port) that test whether it answers, and for a known service **Restart Service**, **Stop Service** and **Journal**.

> **NOTE** `[OK]` only means nothing else is fighting for that port. It doesn't mean the service is healthy. A DVSwitch port clash is a common cause of "no audio". The pairing rules are in `docs/ASL-DVS_Idealized_Node_Configuration.pdf`, section 4.

3.6 Journal tab

The Journal tab lists your pinned services that are installed, followed by any other running service. Click one to open its log in a pop-up window.

CONTROL	WHAT IT DOES
Lines 50 / 200 / All	How far back to read. All means the last 2,000 lines.
grep filter	Shows only lines containing your text, for example `error` or a talkgroup number.
Refresh	Reads the log again.
Restart / Start	Restart or start the service from the log window, so you can watch what happens.
Copy	Copies the lines you see (after the filter).

3.7 Common tasks

A service shows red

1. On the Overview, note the `NR:` count. A high count means it keeps crashing.
2. Open the Journal tab, click the service, choose 200 lines and filter on `error`.
3. Fix the cause. It is usually a config file, covered on the ASL-DVS, DVSM, STFU and M17 tabs (Chapter 4).
4. On the Overview, press **Reset**, then **Start**. Watch the dot turn green.

Find out what is using a port

1. Open the Ports tab and press UDP or TCP.
2. Find the port number. The Process and Known Service columns name the owner.
3. If it shows `[DUP]`, hover over it for both programs, then stop or reconfigure one of them.

Add a service to the Overview

1. Services tab, press **All**, and filter for the service.
2. Click it, pick a group under Pinned Services (or "New group...") and press **Pin**.

Send someone a status report

Use **Copy** on the Overview, the Services list, a Ports table or a journal window, and paste it into your message. Copies are plain text, so they paste cleanly into an email or forum post.

UNDER THE HOOD

How SysMon gathers and acts

Refresh timings

WHAT	HOW OFTEN	NOTES
Background poll (server)	every 10 s	Host, load, memory, swap, temperature, voltage, throttle state, disk, SD health, process count, wifimon shutdown state. Also tells systemd's watchdog that SysMon is alive.
Quick tests, port clashes	every 30 s	Every third background poll.
Status strip (page)	every 5 s	`GET /api/status`, read from the background poll's snapshot, so it costs almost nothing.
Services list	every 30 s	Only while the Services tab is open.
Ports list	every 15 s	Only while the Ports tab is open.
M17 reflector list	hourly check	Downloads `M17Hosts.json` again when it is stale (Chapter 4).

Switching tabs stops the previous tab's timer, so only the tab you're looking at polls.

Where the data comes from

- **Services:** `systemctl show` and `systemctl list-units`. The owner comes from the running process, or from `User=` in the unit.
- **Ports:** `ss -tlnpu`, with each process's command line from `/proc/<pid>/cmdline`.
- **Journal:** `journalctl -u <unit> -n <lines>`, between 10 and 2,000 lines.
- **Power:** `vcgencmd get_throttled`. Its bits give under-voltage, frequency cap, throttling and temperature limit, both now and since boot. Without it, SysMon shows the core voltage instead.

Service actions

Every action goes to `POST /api/svc`. Unit names are checked before anything runs.

- **Simple actions** (start, stop, reset-failed, enable, disable, reload, mask, unmask) run the matching `systemctl` command, with a 15-second limit.
- **Restart** runs in the background: `systemctl stop`, wait up to 15 s; if it is still running, `SIGKILL` and wait up to 5 s; then `systemctl start` and wait up to 10 s for it to become active. The result goes to SysMon's own log.
- **Kill** sends `SIGKILL` to the main process. systemd may then restart it, depending on the unit's `Restart=` setting.
- **Stop All** runs `systemctl stop` on each pinned unit except `sysmon.service`. **Restart All** runs `systemctl restart` on each pinned unit, SysMon included.

The pinned list

The Overview groups come from `[services] pinned` in `/etc/sysmon/sysmon.conf`, one line per entry:

```
__GROUP__:Asterisk Core
asterisk.service | ASL3 Asterisk/app_rpt | /etc/asterisk/rpt.conf | 4569 | udp
# unit | description | config_path | port | proto   ( - for none )
```

You can edit it as text on the Edit tab (Chapter 5). Pin and Unpin rewrite it for you.

Login and process

- SysMon runs as root, because it reads every config file and controls services. It refuses to start as any other user.
- SysMon and the Dashboard keep their logins in one shared file, `/run/asl_dvs/auth_session.json` (mode 600, locked while written). Both use the cookie `asl_dvs_session`, so one login covers both pages. A session expires 12 hours after it was last used. Five failures from one address within 5 minutes lock it out of both for 15 minutes. `/run` is cleared at reboot, so a reboot logs everyone out.
- The password is checked through PAM, with `/etc/shadow` as a fallback.
- The page is built once at start-up and sent compressed. Everything after that is small JSON requests.

```
# SysMon's own checks, without the web page
sudo python3 /usr/local/bin/sysmon.py --selftest
```

Node and DVSwitch health

Five SysMon tabs check the radio side of the node: ASL-DVS (the Asterisk, Allmon3 and DVSwitch config files), DVSM (DVSwitch Mobile accounts), STFU (BrandMeister), M17 (the USRP2M17 bridge) and Phone (autopatch). They read your files and tell you, in plain words, what is set, what is missing and how to fix it.

4.1 How the check tabs read

All five tabs use the same badges, so learn them once:

BADGE	MEANS	WHAT TO DO
PASS / OK	Set, and set to a real value rather than a default placeholder.	Nothing.
WARN	Works, but something is off or optional and missing.	Read the fix line; often it is fine to leave.
FAIL	Missing, still at a default such as `N0CALL`, or wrong.	Follow the fix line under it.
INFO	A value for your reference, such as a port or talkgroup.	Compare it with the other side of the link.
NONE	Listed but turned off (for example a module set to `noload`).	Nothing, unless you need it.

Many check lines end with a **fix**: the exact setting to add or the command to run. Copy buttons on each card give you the whole card as text.

> **CAREFUL** The **Edit** buttons on these tabs open the real config file, and **Save** overwrites it after one confirmation. SysMon writes the file safely (a power cut can't leave half a file), but it **does not keep a backup copy** of these files. Take a restore point in 44helper's Config tab first (Chapter 11), or press Copy before you change anything.

4.2 ASL-DVS tab

Four cards, one per group of files. Click a file row, or its **Edit** button, to open it in the editor.

/etc/asterisk/ — Config Files

Every `.conf` file in `/etc/asterisk`. The ones that matter most are checked under their name:

FILE	CHECKED
rpt.conf	Callsign is set and not a placeholder; each node number; its `rxchannel`, and for USRP nodes the RX and TX ports, with a reminder that Analog_Bridge must use the same pair.
manager.conf	At least one AMI user, and that each has a password.
echolink.conf	Callsign, password, EchoLink node number and the ASL node it links to.
rpt_http_registrations.conf	Each node that registers with AllStarLink, and that it has a password.
savenode.conf	Node and password pairs.
modules.conf	Which `chan_` channel drivers load, are required, or are set to `noload`. This list is folded away: press ▶ **Show**.

extensions.conf	The area code (`HOMENPA`, not the 999 default) and node number in `[globals]`.

Lines in a file that set a port get an `[OK]` or `[DUP]` badge, the same clash check as the Ports tab.

Allmon3 — Config Files

`allmon3.ini`, `web.ini`, `users`, `menu.ini` and `custom.css` in `/etc/allmon3`. `users` is read-only here: change Allmon3 logins with `allmon3-passwd`.

DVSwitch — Config Files

The DVSwitch files that exist on this Pi, from either their `/opt` or `/etc` location:

FILE	CHECKED
Analog_Bridge.ini	DMR ID and repeater ID (not defaults), AMBE mode, TX talkgroup and slot (1 or 2), colour code, software AMBE fallback, and the USRP RX and TX ports.
MMDVM_Bridge.ini	Callsign and DMR ID, location and description, and which modes (DMR, Fusion, P25, NXDN, D-STAR) are enabled, with their addresses. FAIL if no mode is enabled.
DVSwitch.ini	For each mode section: address and ports. For `[STFU]`, also the BrandMeister server, password, DMR ID and start talkgroup.
var.txt	The DMR networks the `dvs` menu knows about (BrandMeister, TGIF, DMR+, Other 1 and 2), and their addresses. Passwords are never shown.
Analog_Reflector.ini	If present: callsign, node ID and the USRP port.

D-Star — ircDDBGateway

Checks `/etc/ircddbgateway` for what the Dashboard's D-STAR gateway linking needs:

- the `ircddbgatewayd` service is running;
- `gatewayCallsign` is set, and `repeaterBand1` is a module letter. The card shows the 8-character name the Dashboard uses for the node, such as `N0CALL B`;
- `ircddbEnabled=1` (FAIL without it) and `dextraEnabled=1` (WARN);
- remote control (optional): when `remoteEnabled=1`, a password is set, the port is valid and something is listening on it;
- `gatewayAddress` is blank or one of this Pi's own addresses, never `127.x` and never the router's address.

Buttons: **Edit /etc/ircddbgateway**, **Restart ircddbgatewayd** (asks first; any D-STAR link drops), **Refresh** and **Copy**.

4.3 DVSM tab

DVSwitch Mobile (DVSM) is the phone app for talking through your node. This tab reads your files and shows exactly what to type into each of the app's four account types. The **Node IP** in the tab header is this Pi's address on your network.

CARD	USE IT FOR	KEY SETTINGS SHOWN
IAX2 Direct	Phone-mode connection straight to your node.	Host (LAN and WAN), port, user name and password from `iax.conf`, CallerID, node. "Phone Mode IAX2" ticked.

IAXRpt	**Recommended.** Radio mode: least data, and it only sends audio while the channel is active.	As above, but Caller Num must be `0` and "Phone Mode IAX2" unticked.
Node Mode	The app registers as its own AllStarLink node.	register.allstarlink.org, port 4569, and a **spare** node number registered to your callsign. Auto Connect unticked.
USRP / DVSwitch Bridge	Connect to Analog_Bridge's USRP port.	Host (local, LAN or WAN), the TX and RX ports, callsign and DMR ID.

Passwords show as `••••••••`. Click the eye to see one while you copy it into the app.

Compatibility Checks

SETTING (RPT.CONF NODE)	WHAT IT DOES	SHOWN AS
propagate_dtmf	Lets DTMF commands from DVSM reach the node.	INFO if off: only needed for DVSM.
phonesendlinks	Fills the DVSM Status tab's linked-node list.	INFO if off.
propagate_phonedtmf	Passes phone callers' tones to every linked node, bridges included.	**WARN if on.** Leave it off unless you need it.
remote_dtmf_allowed	Lets any linked node send commands to this node.	**WARN if on.**

It also checks `iax.conf`: that an `[iaxrpt]` (or callsign-named) user with a password exists, which IAXRpt needs, and how call tokens are set.

4.4 STFU tab

STFU is DVSwitch's BrandMeister terminal: it connects the node to BrandMeister directly, as a hotspot. Its settings live in the `[STFU]` section of `DVSwitch.ini`.

CARD	SHOWS
Installation	Whether the STFU program and its service are installed, running and enabled at boot.
Configuration	BrandMeister server, port (default 62031) and hotspot password (masked); your DMR ID; the USRP ports; start talkgroup; subscriber file; talker alias; log level. A link opens BrandMeister's master-server list.
Compatibility Checks	Server set, password set, DMR ID not a default, ports set, subscriber file found (it turns DMR IDs into callsigns), service running.
Sample [STFU] Stanza	A complete section built from your values, with placeholders for anything missing. **Copy Stanza** or **Edit Stanza**.
Edit DVSwitch.ini	Expand to edit the whole file. **Save**, then **Restart STFU** to apply.

> **TIP** The BrandMeister hotspot password is not your BrandMeister login password. Set it on brandmeister.network under **Self Care → Hotspot Security**.

4.5 M17 tab

The M17 bridge, USRP2M17, links private node 1917 to an M17 reflector. The Dashboard's M17 tab rewrites its settings file every time you connect or disconnect.

CARD	SHOWS
Installation	The program, its service and whether it is running. **Restart usrp2m17** asks first.
Configuration	`/opt/USRP2M17/USRP2M17.ini` : callsign; the reflector name and address (`0.0.0.0` means not connected); the M17 and USRP ports; gain; logging.
Compatibility Checks	Callsign set; reflector connected (WARN when idle, which is normal); USRP ports are 32008 and 34008; service running; `rpt.conf` has a `[1917]` node whose `rxchannel` ports match the ini file.
Reflector Host File	`/var/lib/mmdvm/M17Hosts.json` : how many reflectors it lists and how old it is. **Update Now** downloads a fresh copy.
Sample USRP2M17.ini	A complete file built from your values.
Edit USRP2M17.ini	The real file. Save, then Restart from the Installation card.

CAREFUL Edits to `USRP2M17.ini` don't survive the next M17 connect or disconnect on the Dashboard, which rewrites the whole file. Make lasting M17 changes on the Dashboard instead.

NOTE If `LocalPort` under `[M17 Network]` is missing, the sample fills in `32010` . That is the port MMDVM_Bridge uses for P25. The recommended value is `17010` (see the node configuration guide, section 4).

4.6 Phone tab

The Phone tab shows what Asterisk is set up to do for **autopatch** (radio dials out) and **reverse autopatch** (phone calls in to the radio). It is **read-only**. You set phone networks up on the Dashboard's Edit page, under Phone. The one exception is the Autopatch modules card.

CARD	WHAT IT TELLS YOU
Live	IDLE, DIALING, INCOMING or IN CALL; whether the phone node is linked to your node (it links while the Dashboard Phone tab is open); whether that tab is open (callers get busy when it isn't); which network is in use; phone patch on or off; calls now; sign-in state. Refreshes every 5 seconds.
Autopatch node	The private node the Dashboard made for each phone network.
Phone networks	Each network and its dialing target.
Hams Over IP	Shown only when you have a Hams Over IP account: the account and its sign-in.
Dialing rules	The dialplan contexts Asterisk has for each network.
Health checks	Asterisk running; nodes, files, live state and call routes all consistent. Each check is OK, WARN, FAIL or INFO, with a fix.

Router test	Three tests for calls that won't connect or have one-way audio (below).
Autopatch modules	The Asterisk modules each feature needs, and whether each is loaded. You can change them here (below).
Recent call attempts	Up to 20 phone calls from the last 24 hours, in or out, with the network and the last 4 digits only.

Router test

1. **Going out: Run outgoing test** signs in to your phone provider once and sends one STUN packet. It takes about 5 seconds.
2. **Call audio:** start a **Test call** on the Dashboard's Phone tab, then press **Check audio now**. SysMon counts audio packets in and out over 4 seconds. "Out but none in" points to the router forward or the Pi firewall. Audio in but none out is normal on a simplex node while nobody talks.
3. **Coming in:** press **Start watching (10 min)**, have another Hams Over IP member call your extension, then press **Check**.

Autopatch modules

Modules are grouped by feature: **Autopatch** and **Reverse autopatch**, each READY, MISSING, OFF, NOT SET UP or ASTERISK DOWN. Modules both features need, and the ones every node needs, are locked (🔒).

- **Activate / Deactivate** a feature: sets every module it needs to load and starts it, or sets the modules only it uses to noload and stops them. The confirmation lists exactly which modules.
- Per module: **Load** / **No load** change `modules.conf` for the next Asterisk restart; **Start** / **Stop** load or unload it now.
- If a module file is missing from the Pi, the card names the package it comes from and offers **Reinstall**, with the same commands shown for doing it by hand.

> **NOTE** Unlike the file editors, this card backs up `modules.conf` before every change. It keeps the last 5 copies next to the file.

The full story of the phone dialplan is in `docs/ASL-DVS_Pi02w_Phone_Dialplan_Guide.pdf`.

4.7 Common tasks

No audio between AllStar and DMR

1. ASL-DVS tab: under `rpt.conf`, note node 1999's RX and TX ports.
2. Under `Analog_Bridge.ini`, check that its RX Port and TX Port are the same pair the other way round, as the note says.
3. Ports tab: check that neither port shows `[DUP]`.
4. Fix the file with Edit, then restart `analog_bridge` and `asterisk` on the Overview.

Set up DVSwitch Mobile

1. DVSM tab: make sure the IAXRpt card shows PASS. If not, follow the Compatibility Checks fix.
2. In the app, add an IAX2 account and copy each field from the IAXRpt card. Use Host (LAN) at home and Host (WAN) away.

3. Untick Phone Mode IAX2 and set Caller Num to 0.

Get a new node ready for STFU

1. STFU tab: work down the Compatibility Checks until every line shows PASS.
2. Copy the Sample [STFU] Stanza if your file has no `[STFU]` section, and paste it in with Edit DVSwitch.ini.
3. Save, then Restart STFU.

UNDER THE HOOD

Files, endpoints and what is sent to the page

Where each tab reads

TAB	FILES AND COMMANDS	ENDPOINT
ASL-DVS	every `*.conf` in `/etc/asterisk`; `/etc/allmon3/*`; the DVSwitch files under `/opt`, `/etc` and `/var/lib/dvswitch/dvs/var.txt`; `/etc/ircddbgateway`, plus `ss -Hlunp` and `ip -o -4 addr`	`/api/asterisk/files`, `/api/allmon3/files`, `/api/dvswitch/files`, `/api/dstar-gw`
DVSM	`iax.conf`, `rpt.conf`, Analog_Bridge's ini; the LAN address from the default route	`/api/dvsm`
STFU	`DVSwitch.ini [STFU]`, `/opt/STFU`, `stfu.service`	`/api/stfu`
M17	`/opt/USRP2M17/USRP2M17.ini`, `rpt.conf [1917]`, `usrp2m17.service`, `M17Hosts.json`	`/api/m17`
Phone	Asterisk's files and the CLI (`asterisk -rx`), `/var/log/asterisk/full` or `messages` (last 2 MB), or the journal	`/api/phone`, `/api/phone/routertest`, `/api/phone/modules`

Outside connections

- **DVSM** finds your public (WAN) address from `api.ipify.org`, falling back to `checkip.amazonaws.com`, with a 2-second limit each. Without internet the WAN rows say "Could not detect".
- **M17 host file** comes from `hostfiles.refcheck.radio`, falling back to `m17-project.github.io`. A download is only kept if it is at least 2 KB and lists at least 10 reflectors. The background poll checks every hour and fetches a new copy when the file is over 24 hours old.

Secrets

- The DVSM and STFU cards send the account passwords to your logged-in page, masked until you press the eye, so you can copy them into an app.
- The ASL-DVS checks report only whether a password is set, never its value. `var.txt` password lines are skipped completely.

Writes

- File saves go to `POST /api/asterisk/file`, `/api/dvswitch/file` and `/api/allmon3/file`. Each writes a temporary file, keeps the old file's owner and mode, syncs it to disk, and swaps it in place. No backup is kept. Asterisk file names are validated, and the DVSwitch and Allmon3 editors only accept the files in their fixed lists.
- Autopatch module changes edit only SysMon's own marked block in `modules.conf`, between `; >>> sysmon-phone modules` and `; <<< sysmon-phone <<<`. A conflicting `noload` elsewhere is commented out with `; sysmon-phone off:`. If the Dashboard's own block says noload, SysMon warns instead and leaves it, because Save Phone on the Dashboard would undo the change. Backups are named `modules.conf.sysmon-bak-YYYYMMDD-HHMMSS`, keeping 5. Live start and stop use `module load` / `module unload` through the Asterisk CLI.
- Restarts use `systemctl restart` on `stfu`, `usrp2m17` or `ircddbgatewayd`.

Defaults the checks treat as "not set"

Callsigns `N0CALL`, `NOTSET`, `NONE` or blank; the stock placeholder DMR and repeater IDs (such as `123456789`); the stock STFU server; area code `999`; and M17 address `0.0.0.0`.

System care

The last five SysMon tabs look after the hardware and the Pi itself: Tune (radio audio levels), Hardware (AMBE dongle, audio, power, video), SD Card, Security and Edit (SysMon's own settings). These tabs are where SysMon changes things most, so each section says what it writes and whether it restarts anything.

5.1 Tune tab

For a node with a radio on a USB sound interface run by Asterisk's **SimpleUSB** driver. A bridge-only node with no radio can ignore this tab.

SimpleUSB — Tune Settings

CONTROL	WHAT IT SETS
USB Device	The interface SimpleUSB is using (read-only).
RX Mixer Set	Receive level from the radio into the node, 0 to 1000. Slider or exact number.
TX Mix A Set / TX Mix B Set	Transmit level to the radio on each output channel, 0 to 1000.
Advanced settings	Carrier From and CTCSS From (where COS and tone detect come from: `no`, `usb`, `usbinvert`, `pp`, `ppinvert`); RX Boost; De-emphasis; Pre-emphasis; PL Filter; Invert PTT; Legacy Audio Scaling.
GPIO pins	Extra GPIO lines on the USB interface: in, out0 or out1. Pin 3 is PTT, and pins 5 to 8 exist on CM119 chips only.
Parallel port pins	Folded away: pins 1 to 17 as in, out, PTT, COR or CTCSS.
RX On Delay / TX Off Delay	Tick to turn on, then 0 to 100 frames.

An amber ● **unsaved changes** appears as soon as you move anything. **Reset** puts the card back to what is in the file. **Save** opens a dialog:

- **Save as preset**: also store the three levels as Radio 1 to 5, with a name such as "Mobile" or "Base".
- **Restart Asterisk after saving**: apply now. Without it, the file is saved but Asterisk keeps the old levels until it next restarts.

The Radio 1 to 5 buttons

The Radio buttons in SysMon's command bar (Chapter 3) are your saved presets. Hover over one to see its three levels. A `*` after the name means the levels have been changed since you applied that preset. Empty presets are greyed out, and so is every button when no SimpleUSB driver is loaded.

> **CAREFUL** Pressing a Radio preset button writes its levels to `simpleusb.conf` and **restarts Asterisk at once, with no confirmation**. Every link drops and the node is off the air for a few seconds. The button spins until Asterisk is back. Pressing the preset that is already applied does nothing.

Node Settings — rpt.conf

Folded away by default; press **Show**. It lists every node in `rpt.conf` with its role (radio, USRP bridge, hub, phone node) and four settings:

SETTING	RANGE	MEANING
duplex	0 to 4	How the node handles transmit and receive. 0 is right for a bridge node, 2 for a typical full-duplex repeater.
telemdefault	0 to 2	The default telemetry (voice announcement) mode: 0 off, 1 on, 2 the reduced setting. Bridge nodes use 0.
hangtime	0 to 60000 ms	How long the transmitter stays keyed after a signal drops.
linktolink	yes / no	Link-to-link radio mode.

Each value says whether it is *set on node*, inherited *from* a template, or the *default*. Nodes written by the Dashboard's phone setup, or set in another file, are locked here and say where to change them. Saving changes only the lines that differ, and can reload `app_rpt` so the change takes effect without a full restart.

5.2 Hardware tab

Copy hardware report at the top copies the whole tab as text, ready for a help request.

CARD	WHAT IT SHOWS AND DOES
AMBE Device	A ThumbDV or similar AMBE vocoder dongle, if present. You can **ping** the chip (it answers with its product ID), set the serial port speed (from a fixed list), add or remove a fixed `/dev` name for it with a udev rule, and **reset** it. Reset stops the DVSwitch services, resets the chip, then starts Analog_Bridge, MMDVM_Bridge and md380-emu first and the rest 3 seconds later.
Audio Device	The sound interface the node uses.
Power & Thermal	Supply health from the firmware: under-voltage, throttling and temperature limits, now and since boot.
Video & GPU Memory	Two switches that free memory on a headless Pi (below).
USB Bus	Every USB device, as a **Flat** list or a **Tree**. "possible AMBE" is only a guess from the chip maker. Only a badge with the exact product name, such as ThumbDV, is a confirmed match.
asl-find-sound	ASL's own sound-device finder, for SimpleUSB setup.
ALSA Cards	Linux sound cards, **Playback** or **Capture** side.

Video & GPU Memory

A Pi Zero 2 W used as a node never drives a screen, but the video driver and GPU still hold memory.

- **Disable HDMI and video driver** comments out the `dtoverlay=vc4-kms-v3d` line in `/boot/firmware/config.txt`.
- **Lower GPU memory** sets the GPU's share to 16 MB.

Both take effect after a **reboot**; until then the card marks them as pending. Each can be turned back on the same way.

> **CAREFUL** With the video driver off, a monitor on the HDMI port stays blank, and HDMI sound disappears. If any config file uses a sound card by number, the numbers can shift when the HDMI sound card goes away. When that is a risk, the card lists those lines so you can check them before you reboot.

> **NOTE** These two settings survive an instmon update. SysMon saves them when it is uninstalled for the update, and puts them back if it is reinstalled within 30 minutes (Chapter 2).

5.3 SD Card tab

A Pi runs from its SD card, and SD cards wear out. This tab tells you how the card is doing.

SD Card Health

SECTION	SHOWS
Warnings	Any problem found, in red at the top. The header reads "healthy" or the number of issues.
Device / Identity	The disk, maker, product, serial, manufacture date, firmware and hardware revision, erase size. On eMMC storage, also its own life-time estimate.
Capacity	Card size, and filesystem used and free. Amber from 75 %, red from 90 %. Also inodes used.
Filesystem (ext4)	State (should be `clean`), mount count, last check, lifetime writes, what happens on errors.
I/O counters	Bytes read and written since boot. This is a guide to wear, not a lifetime total.
Errors	Whether the root filesystem has dropped to **READ-ONLY** (usually corruption), and how many storage errors the kernel has logged, with the last lines.

Integrity Tests

Run fsck -n (advisory) checks the filesystem without changing anything, with a progress bar and a **Cancel** button. The result is only advisory because the card is in use while it runs. If you leave the tab, the test keeps running in the background.

> **TIP** To make the card last longer, use 44helper's System Optimization tab (Chapter 10). To keep a spare, take an image with instmon's Disk Image section (Chapter 13).

5.4 Security tab

14 passive checks of the node's security. The bar at the top counts Pass, Warn and Fail, with **Re-run checks** and **Copy**. The checks are grouped in four cards:

CARD	CHECK	SEVERITY
DVSwitch	BrandMeister hotspot password set	medium
AllStarLink	AMI port not reachable from the network	high
	AMI has no default or weak password	high

	IAX2 secret strength	medium
	Legacy `chan_sip` module disabled	low
	`/etc/asterisk` permissions	low
usrp2m17	USRP bridge ports bound to 127.0.0.1 only	high
	M17 gateway port exposure	high
Cross-cutting	SSH hardening	medium
	fail2ban jails cover the exposed services	medium
	Monit has authentication	medium
	Dashboard companion app requires a login	high
	Services run as non-root where possible	medium
	Service files and programs not writable by others	high

Click a check to open its explanation: **what it reads**, **what the result means**, **why it matters**, **how to fix it** and references. The amber Owner and Perms values on the Overview jump straight to the matching explanation here.

> **NOTE** The checks only read config files, `/sys`, `/proc`, `ss` and systemd. They never connect to AMI, M17, USRP or the Dashboard, so running them can't trigger fail2ban or fill a security log. Checks that read passwords report only yes or no.

> **CAREFUL** Take a FAIL on the USRP port check seriously. USRP has no password: anyone who can reach that port can key your transmitter under your callsign. Bind it to `127.0.0.1`.

5.5 Edit tab

SysMon's own settings. Change the fields, then press **Save** in the bar at the bottom of the page. **Reload from file** discards your changes.

SECTION	WHAT IT HOLDS
Visible Tabs	Tick the tabs to show. Overview and Edit are always on. A hidden tab stops polling, though an SD card test already running carries on.
Identity	Callsign, node and label for SysMon's own use.
Server	SysMon's port (9999) and bind address (0.0.0.0, meaning every interface). A change needs a SysMon restart, and a banner says so.
Thresholds	**NRestart warn** (3) and **NRestart crit** (10) colour the restart badge in the service panel. **CPU warn %** and **RSS warn MiB** are stored but not used by this version.
ASL-DVS Config Files	Folded away. Every Asterisk, DVSwitch and Allmon3 file with **Edit**, and **Hide** / **Show** to keep files you never touch off the ASL-DVS tab.
Pinned Services	Folded away. The Overview list as text: `unit | description | config_path | port | proto`, groups as `__GROUP__:Name`.

The bar also has four tools:

- **ASL AllConf** shows every Asterisk and Allmon3 file in one window, each under a header, with Copy All. It is ideal for sending your whole setup to someone helping you. **DVS AllConf** does the same for the DVSwitch files.
- **asl_dvs.conf** and **sysmon.conf** open the Dashboard's and SysMon's own config files in the editor.

> **CAREFUL** AllConf output includes passwords and secrets from your files. Remove them before you post it anywhere public.

5.6 Common tasks

Set and save radio levels

1. Tune tab: move RX Mixer Set while someone talks to you, until their audio sounds right on Allmon3 or a linked node.
2. Set TX Mix A (and B if used) for correct deviation on your radio.
3. Save, pick a preset slot and name, and tick Restart Asterisk.
4. From then on, one press of that Radio button brings these levels back.

Free memory on a Pi Zero 2 W

1. Hardware tab, Video & GPU Memory: turn on both options. Check any sound-card lines it lists.
2. Reboot from the command bar. Afterwards the card should no longer show "pending".

Is my SD card failing?

1. SD Card tab: look for red warnings, READ-ONLY, or kernel storage errors.
2. Run fsck -n. Errors in the result, along with kernel errors, mean it's time to image the card and replace it.

UNDER THE HOOD

What each tab writes

ACTION	FILE OR COMMAND	BACKUP
Tune Save	`/etc/asterisk/simpleusb.conf`; values range-checked; levels mirrored to `[radio_tune]` in `sysmon.conf`	none
Save as preset	`[radio_presets] radioN_values / radioN_title` in `sysmon.conf`	—
Restart Asterisk / preset apply	`asterisk -rx "module unload chan_simpleusb.so"` (allowed to fail), 2 s pause, `systemctl restart asterisk` (30 s limit). One restart at a time; the page polls `/api/radio/stack_status` until `core show uptime` answers.	—
Node Settings Save	Only changed keys inside the node's own `[node]` section in `rpt.conf`, comments kept; optional `module reload app_rpt`	none
Video / GPU memory	`/boot/firmware/config.txt`: tags the vc4 overlay line with `#ASL-DVS-VIDEO-OFF#`; adds `[all] gpu_mem=16` between `#ASL-DVS-GPUMEM-BEGIN` and `-END`	`config.txt.asl_dvs.bak` on every write
AMBE udev name	`/etc/udev/rules.d/99-ambe-sysmon.rules` (USB IDs: 4 hex digits; name: letters, digits, `_` and `-`, up to 32)	—
Edit Save	`/etc/sysmon/sysmon.conf` (identity, server, thresholds, pinned list, visible tabs, hidden files)	—

Reads

- **Power:** `vcgencmd get_throttled`. **Video:** `/proc/modules` (vc4), `/sys/class/drm/card*-HDMI-*/status`, `CmaTotal` in `/proc/meminfo`, `vcgencmd get_mem gpu` and `/proc/asound/cards`. A setting is "pending" when `config.txt` and the live state disagree.
- **SD card:** the root device's identity from `/sys/block`, `df`, `tune2fs -l` (needs root), the kernel I/O counters, the root mount options, and storage lines from the kernel log. The test runs `fsck -n` in the background; the page polls it every 1.5 s.
- **Security:** each check lists its exact source in its explanation. The same checks run from the shell with `sysmon.py --selftest`.

Boot-option carry-over on update

`--uninstall` records which boot options were on in `/etc/sysmon/boot_restore.json` (mode 600) and restores the stock `config.txt`. `--install` re-applies them if that record is under 30 minutes old, then deletes it. When updating from a SysMon too old to write the record, a fresh install falls back to `config.txt.asl_dvs.bak`, but only if that backup is under 30 minutes old and the service file is gone.

Full-build tabs

The full SysMon build, for a Pi 3, 4 or 5, adds five tabs the Pi Zero 2 W build leaves out to save memory: Firewall, Reg, Net, Zello and Console. Everything in Chapters 3 to 5 also applies to the full build, which has the same tabs.

FULL BUILD This chapter covers `sysmon_v6_13_74_20261007.py` (version `6.13.74`). On a Pi Zero 2 W you can install the full build with instmon's **Full build (override)** choice, but it uses about twice the memory.

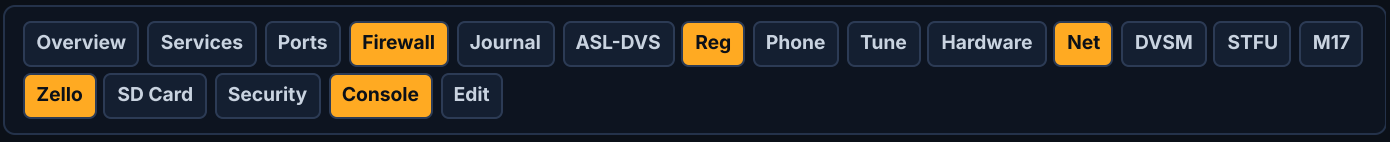


Figure 6.1 · Illustration. The full build's 19 tabs. The five highlighted are covered here.

6.1 Firewall tab

Shows and edits the Pi's own firewall. The **Backend** bar says which firewall SysMon found (ufw, firewalld, nftables or iptables) and whether it is active. A **Guide** explains ufw, firewalld zones and ASL3's default posture in plain words.

PART	WHAT IT DOES
Rule table	Number, port, action (ALLOW, DENY, LIMIT), source and the known service for each rule. Click a rule for details and **Delete**, which asks first. Deleting a firewalld service warns you if it is SSH or Cockpit.
Add Rule	Port, protocol (tcp, udp, any), action and an optional source address, then **+ Add**.
Zone details	For firewalld: which zone each interface is in. A port only opens if it is allowed in the zone of the interface traffic arrives on.
Raw firewall output	The full listing, including rules the table can't show (rich rules), with Copy.
Enable / Disable ufw	Turn the whole ufw firewall on or off. Disable asks first.

CAREFUL **Enable ufw acts at once, with no confirmation, and adds no rules for you.** If there is no rule allowing SSH (22/tcp) and the web pages (8989, 9999, 8990, 8991, 9997) from your network, turning ufw on can lock you out of the Pi. Add those ALLOW rules *first*, and add IAX2 (4569/udp) so AllStar links keep working.

On firewalld, SysMon adds each rule both live and permanently, so it survives a reload or reboot. 44helper's Firewall tab (Chapter 7) has a fuller zone-based editor with built-in guardrails, and works on both builds.

6.2 Reg tab

Checks how your node registers with AllStarLink, which is how other nodes find it.

- **HTTP Registration** reads `rpt_http_registrations.conf`; **IAX Registration** reads the `register =>` lines in `iax.conf`. For each: node numbers valid, a password present, the line syntax correct, and the live state from

Asterisk (registered or not).

- **Dual registration** warns when both HTTP and IAX are active. Pick one and comment out the other; HTTP is ASL3's default.
- **Re-check** reads it all again. Live state also refreshes in the background every 30 seconds.

6.3 Net tab

CARD	WHAT IT DOES
Hostname & name announcer	Three steps, each with its command shown, **Run** and **Copy**: (1) set the hostname (letters, numbers and dashes, up to 63 characters); (2) install and start `avahi-daemon`, so the node answers to `name.local`; (3) test it. The IP address keeps working. You log in once more for each new name, and new backup and image file names use it. AllStar and DVSwitch settings aren't touched, and no reboot is needed.
WiFi	The current WiFi connection and signal.
Install mtr	Shown only if `mtr` is missing: installs `mtr-tiny`.
mtr trace	A 5-cycle route trace to a choice of targets, showing where along the path packets are lost.
Ping	5 pings to Google (8.8.8.8), allstarlink.org, or your DVSwitch servers.
Reflectors	Ping or trace any reflector configured on the Dashboard.

Only one job runs at a time; **Stop** ends it. Output appears in the card's console box.

6.4 Zello tab

For `asl-zello-bridge`, which links private node 1918 to a Zello channel. It is laid out like the STFU and M17 tabs (Chapter 4):

CARD	SHOWS
Installation	The bridge program (`/opt/asl-zello-bridge`) and its service.
Live Status	Read from the journal, not just whether the process is up: who keyed and unkeyed, and when. **Full Journal** opens the service panel.
Configuration	The settings in the service's systemd override (Free mode): Zello account and channel, USRP addresses, ports and gain.
Compatibility Checks	USRP ports 34012 / 32012, a `[1918]` node in `rpt.conf` that matches them, and credentials present.
Sample Override / Edit Override	A reference copy with placeholders, and the real file with **Start**, **Stop**, **Restart**, **Save** and **Copy**.

> **CAREFUL** The override holds your Zello password in plain text, and the editor shows it. Save writes it at once; Restart applies it.

6.5 Console tab

A live Asterisk console (`asterisk -rvvv`) in the browser, plus one-click buttons for the commands you'd otherwise type.

Switching it on and opening it

1. The console starts **switched off**. Press **Enable console** and enter the root password.
2. Press **Open console** and enter the root password again. You need it every time you open the console.
3. Type Asterisk CLI commands as usual. **Copy** and **Clear** work on the screen. **Close** ends the session.

- **Home network only.** It refuses any request that came through a tunnel or proxy (Cloudflare and the like), or from an address outside your LAN.
- **Never a shell.** The `!` key, which runs shell commands from the Asterisk CLI, is blocked, and so is Ctrl-Z.
- Only one console at a time. It closes after 10 minutes with no typing.
- **Disable console** switches it off again, and needs no password.

Command buttons

Pick a node in the **Node** list, then press a button; the result shows below, often with a plain-words summary.

GROUP	BUTTONS
Nodes	rpt stats, rpt lstats, node variables, link details, local nodes, registrations, **Disconnect all links**†
Phone	SIP registrations, endpoints, contacts, **Retry sign-in**†, SIP log for provider, transports, all calls, call quality, dialplan check, phone settings in Asterisk, phone nodes, phone node clash check, **Hang up all calls**†
AllStar links	IAX2 peers, registry, channels, link quality
Audio and CPU	codec conversions; audio (RTP) debug on/off; SIP logging on/off
Detail level	verbose 3, verbose 5, debug off
Reload	**Dialplan reload**†, **PJSIP reload**†
Memory	Swapping now?, programs in swap
System	uptime, version, modules check, **Restart Asterisk when idle**†

† Buttons that change something ask you to confirm. Debug and logging switch themselves off again after 5 minutes. The **Troubleshooting** card below the buttons runs read-only checks and explains the results.

UNDER THE HOOD

How the full-build tabs work

Firewall

- Backend detection order: ufw, firewalld, then nftables or iptables (raw listing only). ufw rules come from `ufw status numbered`; each row's number is what `ufw delete` removes.
- Add runs `ufw allow|deny|limit <port>/<proto>`, with `from <addr>` if given. On firewalld, SysMon runs `firewall-cmd --add-port` both with and without `--permanent`.
- Enable runs `ufw --force enable`; Disable runs `ufw disable`. Endpoint: `POST /api/firewall`.

Reg

Live state: `asterisk -rx "rpt show registrations"`, part of the background poll (every 30 s), with node numbers matched against the config files. Files: `rpt_http_registrations.conf`, `iax.conf`, `rpt.conf` and Allmon3's `allmon3.ini`.

Net

The hostname steps run `hostnamectl set-hostname` and update the `127.0.1.1` line in `/etc/hosts`, then install and enable `avahi-daemon` and `libnss-mdns`, and test the `.local` name from the node itself. Every command is shown before it runs, and Copy gives it to you for SSH if SysMon isn't running as root.

Zello

Service `asl-zello-bridge`; settings in `/etc/systemd/system/asl-zello-bridge.service.d/override.conf` as `Environment=` lines (`USRP_BIND`, `USRP_HOST`, `USRP_RXPORT`, `USRP_TXPORT`, gains and the Zello keys). Expected ports: RX 34012, TX 32012, node 1918. Live status parses `Keyed:` and `UnKeyed:` lines from the journal.

Console

- The on/off switch is the file `/etc/sysmon/console.enable`. It must be a plain file owned by root and not writable by anyone else, or the console stays off. The page writes who switched it on, from where and when. You can also create or remove it from a shell.
- Opening gives a one-time ticket, valid 60 seconds and tied to your login. The console then runs in a real terminal over a WebSocket, with its home directory in `/var/lib/sysmon/console-home`.
- Refused when the request carries proxy headers (`Cf-Connecting-Ip`, `Cf-Ray`, `X-Forwarded-For` and similar), or comes from an address that isn't loopback, link-local or private.
- The command buttons run fixed commands from an allow-list (`/api/console/run`). They don't need the console switched on or open. The node number comes from the list of your local nodes.

PART III

44helper

44helper is the node's setup and repair helper. It installs and removes the radio software, sets up remote access over 44Net or Cloudflare, runs system updates safely, trims SD-card wear, and saves and restores your configuration. SysMon tells you what is wrong; 44helper is usually where you fix it.

7. 44helper basics
8. 44Net and remote access
9. Software installers
.0. Keeping the system healthy
.1. The Config tab

44helper basics

How 44helper is laid out, how its step cards work, and the five status tabs: Overview, Services, Ports, Firewall and Actions Log. Open it at `http://<pi-address>:9997`. It is the same on every Pi.

7.1 The page

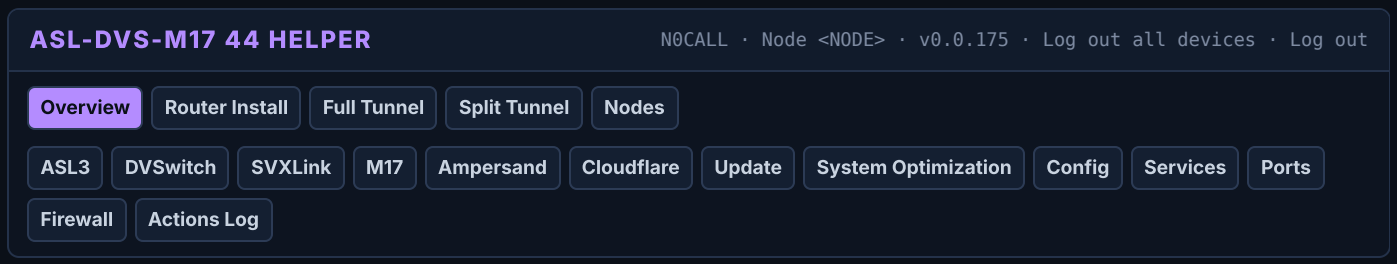


Figure 7.1 · Illustration. 44helper's header and tabs. Row one is 44Net remote access; row two is installers, maintenance, Config and the status tabs.

TAB GROUP	TABS	CHAPTER
Status	Overview, Services, Ports, Firewall, Actions Log	7
Remote access	Router Install, Full Tunnel, Split Tunnel, Nodes, Cloudflare	8
Installers	ASL3, DVSwitch, SVXLink, M17, Ampersand	9
Maintenance	Update, System Optimization	10
Configuration	Config	11

Logging in and out

- Log in with the root password. 44helper has its own login, separate from the Dashboard and SysMon.
- **Your login doesn't time out**, because install jobs can run for a long time. It ends when you press **Log out**, when anyone presses **Log out all devices** (which ends every 44helper login, on every browser), or when the Pi reboots.
- If 44helper restarts while you are logged in, the page reconnects to any running job by itself.

> **CAREFUL** 44helper runs as root and can install, remove and reconfigure anything on the Pi. Keep its port (9997) on your own network. Its Overview warns you if it ever becomes reachable through the 44Net firewall zone.

7.2 How step cards work

Most 44helper tabs are a list of numbered **step cards**. Each card does one job and shows exactly what it will run:

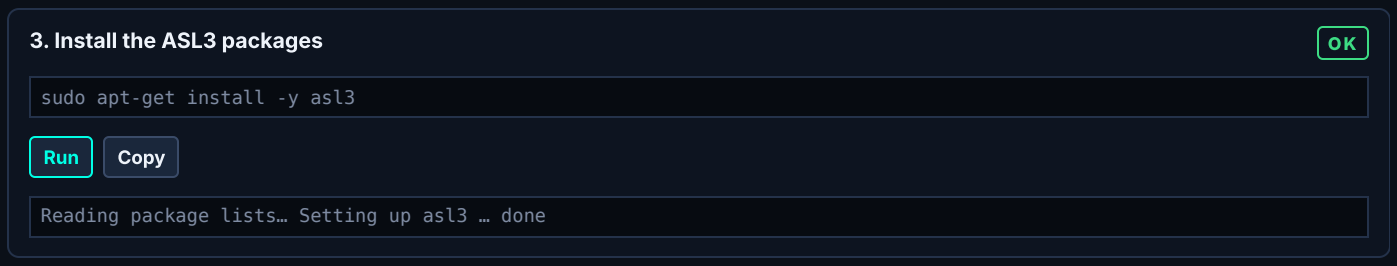


Figure 7.2 · Illustration. A step card: title and status, the command, Run and Copy, and the output.

PART	MEANING
Status pill	**Not run**, **Running…** (with elapsed time), **OK** or **Failed**. **Manual** means you run it yourself over SSH; **Config** means it saves a restore point.
Command box	The exact command. You can edit it before pressing Run, for example to change a version number. Manual steps are read-only.
Run	Runs the command on the Pi, as root, and streams the output below.
Copy	Copies the command (and its output) so you can run it over SSH or send it to someone.
▼ note	A folded note at the top of each tab, and on some cards. Open it before a first run: it says what the step changes and what can go wrong.

- Run the steps **in order**. Each one assumes the ones before it worked.
- Every install tab has a **Script** list (for example Bookworm or Trixie) and a **Custom** choice for typing any command.
- **Purge previous install** shows the removal steps. After a double confirmation they run one after another and stop at the first failure. Destructive steps, such as removing `/etc/asterisk`, are optional and only run if you press their own Run button.
- Step results last only while the page is open; a reload starts the cards fresh. A job that is still running carries on, and the card picks it up again.

> **CAREFUL** Whatever is in the command box runs as root, edits included. Only change a command when you know what the change does.

Safety checks before an install step

- **Masked services.** At the top of each install tab (and the Update tab) a card checks whether any service the tab needs is *masked* (blocked from starting). 44helper clears masks by itself and leaves those services **off**. You can **Turn on** each one, or **Put the mask back (on purpose)** so 44helper leaves it alone. While a mask can't be cleared, the card turns red and the tab's install buttons are locked.
- **Broken packages.** Before any step that installs or removes packages, 44helper asks the package system for unfinished or broken packages. If there are any, the step is refused with "NOT RUN" and a pointer to the Update tab's Package recovery steps (Chapter 10).
- **What to do next.** When an install step fails, a box opens under it with numbered repair steps. Each one has a button that jumps to the right Update-tab card, and a link brings you back.
- **Repair, then purge.** If a purge is refused because packages are broken, it offers to repair them in quiet mode and then run the purge again by itself.

The built-in terminal

A few steps need you to answer questions on screen, for example `asl-menu`, the USRP2M17 installer and Ampersand's first run. Those cards can open a real terminal in the page, but only:

- after the terminal has been **switched on** (from the page with the root password, or by creating `/etc/44helper/terminal.enable` over SSH);
- from your local network, never through a tunnel or Cloudflare;
- with the root password entered again each time you open it;
- for the one command printed on that card, nothing else.

One terminal at a time. It closes after 10 minutes with no typing, or when you log out.

7.3 Overview tab

The banner at the top sums up the worst finding. The cards below it:

CARD	SHOWS
Installed software	ASL3, DVSwitch, USRP2M17, SVXLink and Ampersand. Each is **installed** (with version and running services), **broken** (half-installed packages, or a package present without its program files), **leftovers** (removed but with settings, folders or unit files left behind), or **absent**. Click a name to go to its tab.
Tunnel / wg0	Whether the WireGuard interface `wg0` is up, its address and peer, and the last handshake.
Configured tunnels	The 44Net tunnels 44helper has set up (Pi-hosted or router).
44NetConnect firewall zone	The services and ports open to the 44Net side.
Self-check findings	Red if 44helper's own page, SysMon, the Dashboard, Cockpit or Asterisk AMI is open in the 44NetConnect zone. Amber if `wg0.conf` (which holds your private key) is not mode 600.

Re-check reads everything again straight away; otherwise the installed-software list is cached for 20 seconds.

7.4 Services tab

A grouped list of what is running: System (firewalld, systemd-resolved), 44Net Tunnel, AllStarLink / DVSwitch (Asterisk, the bridges, gateway command ports), M17, SVXLink, Ampersand, Cloudflare Tunnel, and **Dashboards (should stay private)**. Each line is a service or a listening port. Masked services are marked.

7.5 Ports tab

A reference table of every port the node's software uses, checked against what is actually reachable through your remote-access setup. The line above the table says where it looked: the Pi's 44NetConnect zone, the router's forward table, or "none (no tunnel configured yet)".

VERDICT	MEANS
OK	As it should be.
over-exposed	A port that must never be public is open: a dashboard, Cockpit, AMI, a loopback-only bridge port, or a gateway command port.

needed-but-not-open	A port other nodes must reach (IAX2 4569, EchoLink 5198-5199, VOTER 1667, Ampersand hub 4568) is running but not open, so inbound links will fail.
not-listening	Nothing is running on it, which is normal for software you don't use.

7.6 Firewall tab

44helper uses **firewalld**. The guardrail at the top warns when `wg0` is attached to the ASL3 image's own `allstarlink` zone instead of 44helper's `44NetConnect` zone.

CARD	WHAT IT DOES
Pi: 44NetConnect zone	What the 44Net side can reach. Add or remove the services `iax2`, `echolink` and `rtcm`, or a port such as `14569/udp`.
Quick port control (any zone)	Pick any zone (the active and default zones are marked) and **Show zone config**. Presets fill in IAX2, EchoLink, VOTER/RTCM or your DVSwitch Mobile port. Then: **Add (runtime test)** opens it until the next reload, a safe way to try it; **Make permanent** keeps that one port; **Remove** closes it everywhere; **Commit all runtime changes** keeps every test at once; **Reload firewalld**. An advanced section adds services and moves interfaces between zones.
Panic mode	If firewalld's panic mode (all traffic dropped) is on, a button turns it off.
Router: live port-forward table	With router access set up (Chapter 8), the router's forwards, each with a remove button.

As you type a port, a guardrail line checks it: a loopback-only port is **blocked** in every zone, and a dashboard, AMI or Cockpit port is allowed but flagged red.

> **TIP** Use **Add (runtime test)** first. If you lose access, a firewalld reload or a reboot puts things back. Make it permanent once you know it works.

7.7 Actions Log tab

- **Session log**: every command 44helper ran, with OK or FAIL, since it last started. **Refresh** reads it again.
- **Known-good router notes**: saved notes about router setups that worked. Click one to read it.

7.8 Common tasks

Check the node is safe before going on 44Net

1. Overview: the Self-check findings card should show no red lines.
2. Ports: no row should say **over-exposed**. IAX2 should be OK, not needed-but-not-open.
3. Firewall: the guardrail should be quiet, and the 44NetConnect zone should list only what you mean to open.

An install step says "NOT RUN"

1. Read the reason in its output: masked services, or broken packages.
2. For masks, use the card at the top of the tab. For broken packages, follow the "What to do next" buttons to the Update tab.
3. Come back with the link at the top of the Update tab and run the step again.

UNDER THE HOOD

How 44helper runs things

Files

PATH	HOLDS
`/opt/44helper/asl_dvs_m17_44helper.py`	the program (service `44helper`, port 9997)
`/etc/44helper/44helper.conf`	settings: server, identity, tunnels, router access
`/etc/44helper/44helper_guard_keep.json`	services you chose to keep masked
`/etc/44helper/terminal.enable`	terminal switch: must be owned by root and not writable by others
`/run/44helper/sessions.json`	logins (mode 600, on tmpfs, so cleared at reboot)
`/etc/wireguard/wg0.conf`	the 44Net tunnel config (mode 600)

Jobs

- A step's Run posts `{mode, distro, step_id, command_text}` to its tab's `/api/<tab>/action`. The step must exist in that tab's script list; the command text itself runs through the shell, as root.
- **Package-changing commands** (apt, dpkg) run as their own systemd job, not inside 44helper. They keep going if 44helper restarts, and the page re-attaches to them. Package, build and install-script commands have no time limit; anything else stops after 15 minutes unless the step sets a different limit.
- Before a package command: an install step on a tab runs the masked-service check, which unmasks and disables any masked service the tab needs; then `dpkg --audit` must come back empty.
- The page polls `/api/action_status?job_key=…` about every half second, and keeps retrying if the connection drops.
- Every POST needs the login cookie and an `X-Requested-With` header (CSRF protection). Wrong passwords share one 5-try lockout across the login, the terminal and other password prompts.

Port roles

Each reference port has a role. **peer-inbound** ports (4569, 5198-5199, 1667 and 4568, all UDP) should be open when used. **mgmt-LAN** ports (5038, 8080, 8989, 9090, 9997 and 9999) and **loopback-only** ports (31100, 31103, 32008, 34008, 32012, 34012, and the command ports 6073-6075 and 54321) must never be exposed. **outbound-only** (7844, Cloudflare) never needs an opening. "Exposed" means in the 44NetConnect zone (Pi-hosted), in the router's forwards (router, single address), or the whole subnet (router, routed subnet with masquerade off).